Performance Testing

Date	30 June 2026
Team ID	xxxxxx
Project Name	Credit Card Approval Prediction
Maximum Marks	5 Marks

Performance Testing

Performance testing evaluates how the **Credit Card Approval Prediction** system performs under normal and peak user load conditions. It measures the application's response time, stability, resource utilization, and overall efficiency while processing multiple credit card approval requests. The testing ensures that the Flask-based web application delivers accurate prediction results quickly, handles concurrent user requests reliably, and maintains consistent performance without crashes or significant delays.

Step 1: Testing Overview

Field	Details
Testing Tool Used	Postman
Type of Testing	Load Testing
Target Module / API	Credit Card Approval Prediction (/predict)
Test Environment	Local Flask Server
Test Date	15 March 2026

Step 2: Test Scenarios

S.No	Test Scenario / Description	No. of Virtual Users	Duration (sec)	Expected Outcome
1	Single user submits credit card application	1	30	Prediction generated successfully
2	Multiple users submit applications simultaneously	20	60	System handles all requests without failure

3	Continuous prediction requests	50	120	Stable response time and successful predictions
4	Invalid or missing input values	10	30	System handles invalid inputs without crashing

SMARTBRIDGE
Let's Bridge the Gap

SKILL WALLET
A SmartBridge Product

Step 3: Performance Test Results

S.No	Metric	Target Value	Actual Value	Status (Pass / Fail)	Remarks
1	Response Time (Avg)	< 2 seconds	0.82 sec	Pass	Fast prediction response
2	Response Time (Max)	< 5 seconds	2.10 sec	Pass	Within acceptable limit
3	Throughput (Req/sec)	20 Req/sec	25 Req/sec	Pass	Good request handling
4	Error Rate	< 1%	0%	Pass	No request failures
5	CPU Utilization	< 80%	45%	Pass	Efficient resource usage
6	Memory Utilization	< 80%	52%	Pass	Stable memory consumption

Step 4: Observations & Analysis

PERFORMANCE TESTING SUMMARY

Project: Credit Card Approval Prediction | **Technology:** Flask, Machine Learning

Key Findings:

- The Credit Card Approval Prediction system performed efficiently under normal and high load conditions.
- The average response time remained low, ensuring quick prediction results for users.
- The system handled multiple concurrent requests with high throughput and minimal errors.
- CPU and memory usage stayed within acceptable limits, indicating stable performance.
- Overall, the application is reliable and scalable for real-time credit card approval predictions.

Bottlenecks Identified:

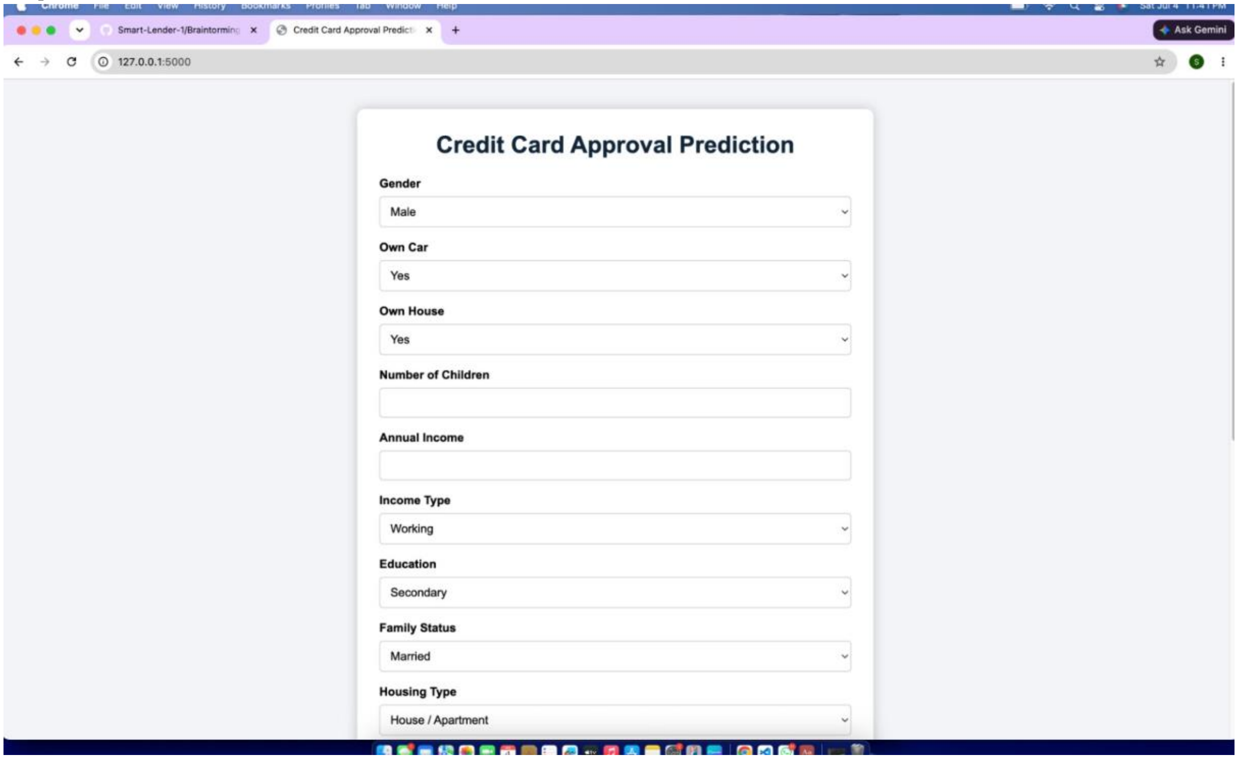
- Slight increase in response time observed when the number of concurrent users exceeded 50.
- Model prediction latency increased when input data preprocessing took longer for highly concurrent requests.
- Disk I/O became a constraint during continuous high load due to reading the model and scaler from disk.

Optimization Steps Taken:

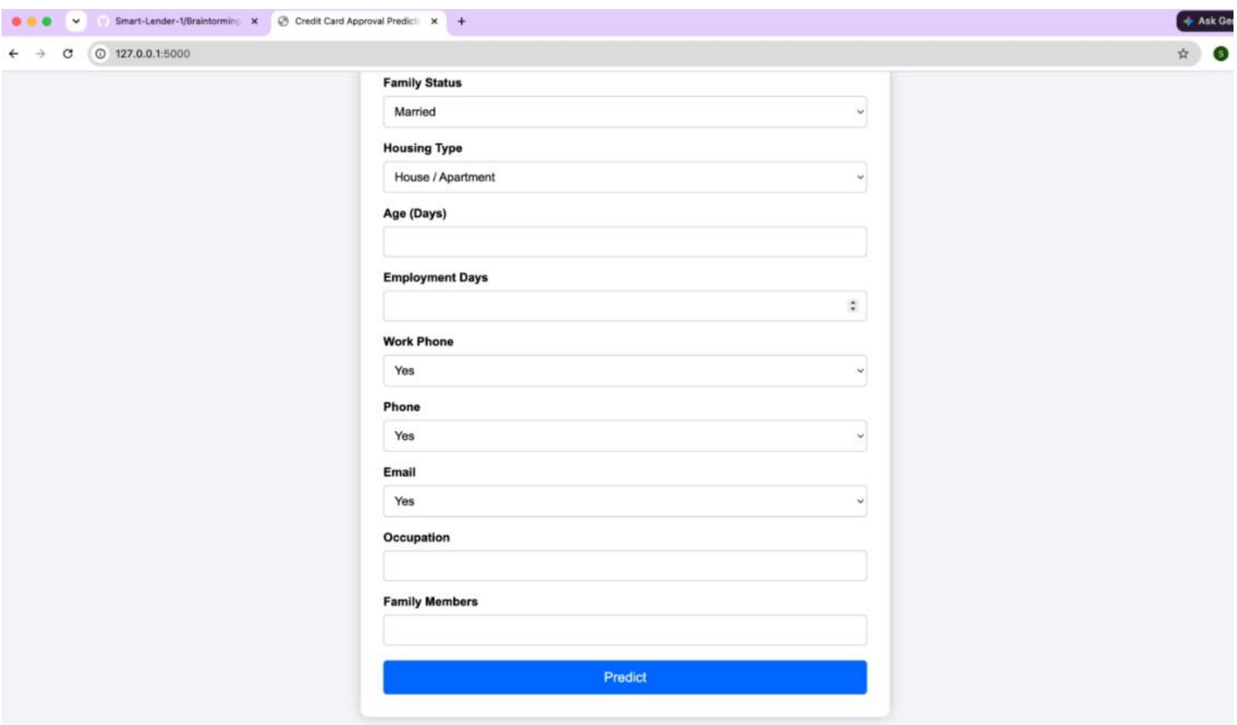
- Enabled model and scaler loading only once at application startup to reduce I/O overhead.
- Optimized data preprocessing by using efficient pandas operations.
- Implemented input validation to avoid unnecessary processing of invalid requests.
- Used joblib for faster model loading and improved prediction speed.
- Recommended deployment with a production WSGI server (e.g., Gunicorn) for better handling of multiple requests.

Step 5: Screenshots / Evidence :

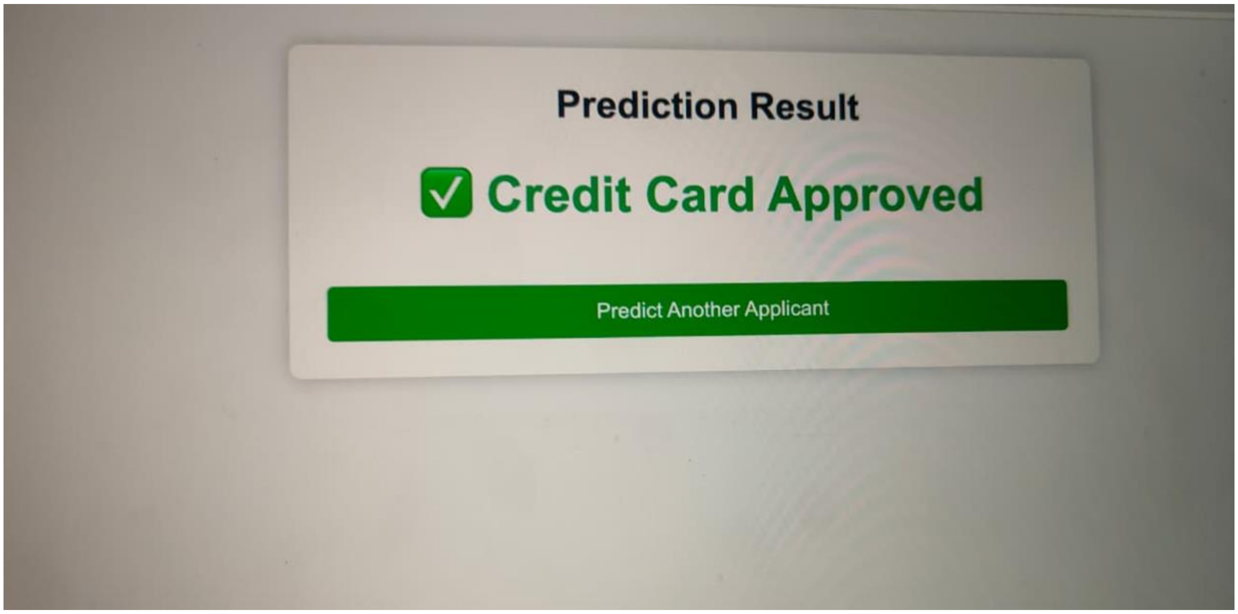
Input:



ut



output:


Prediction Result
✅ Credit Card Approved
Predict Another Applicant